Grokking Under a Fixed Budget

One Long Run or Several Restarts?

Probability and Statistics | BAMAT207
Final study | 15 October 2026

Name: [Your name]
Registration number: [Your registration number]

The problem: delayed learning on new questions

Grokking: a delayed jump from practice answers to new questions

Memorization: getting familiar practice questions right.

Generalization: using the rule on questions held back from practice.

Grokking: new-question performance improves after practice performance.

Our objectives: measure the waiting time, compare settings, and compare equal training budgets.

Like learning a maths rule after practising examples; the analogy does not claim human-like understanding.

A controlled experiment with 120 main attempts

One fixed arithmetic rule, repeated independent starting points

Task: (30 + 4) mod 31 = 3; “mod” keeps the remainder.

961 pairs: 40% uses 384 practice + 577 new; 60% uses 576 + 385.

Weight decay gently shrinks model weights; compare 0.3, 1 and 3.

20 starting seeds × six settings = 120 main attempts; 9 pilot runs are separate.

Python trains and records; R calculates the statistical summaries.

A seed chooses repeatable starting weights. An update is one learning step.

One clear rule for success

We use the same success rule for every attempt

T = first update with 3 checks in a row at ≥99% practice and ≥95% held-out accuracy.

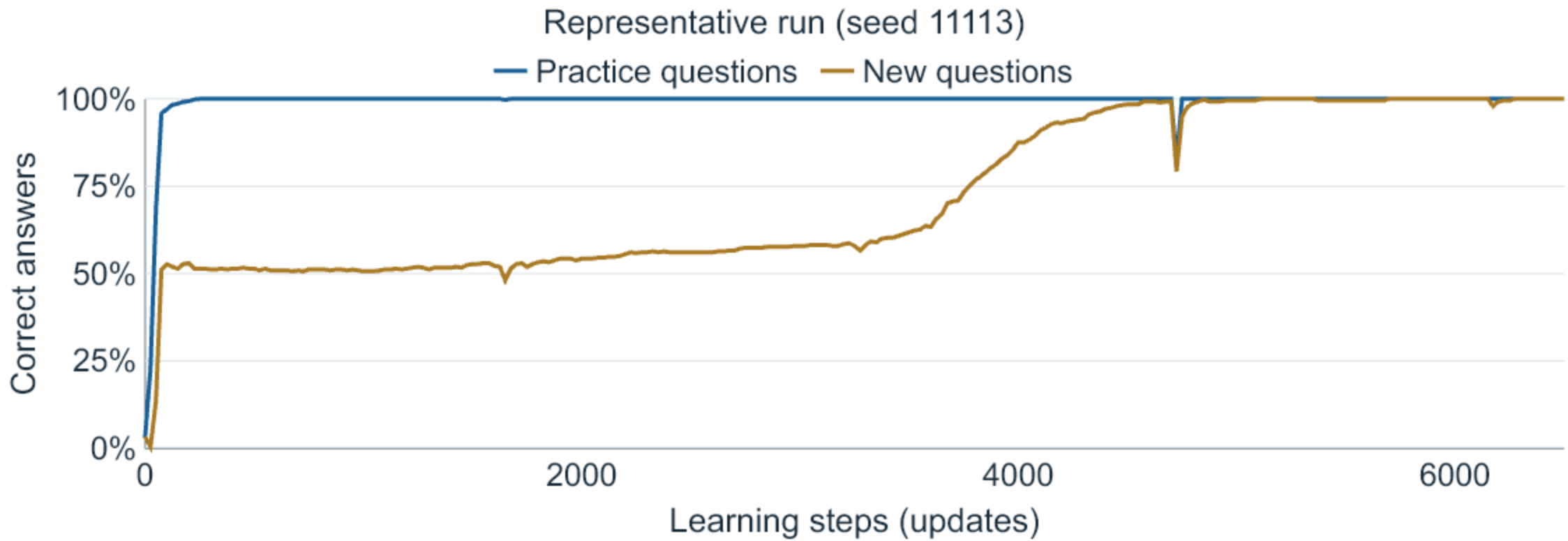
Checks occur every 25 updates. The limit is C = 30,000 updates.

If no event is seen, eventual T is unknown. For observed comparisons: U = min(T, C).

37 of 120 main attempts had no confirmed event by the limit. “Run completed” ≠ “generalized.”

The learning delay in one example

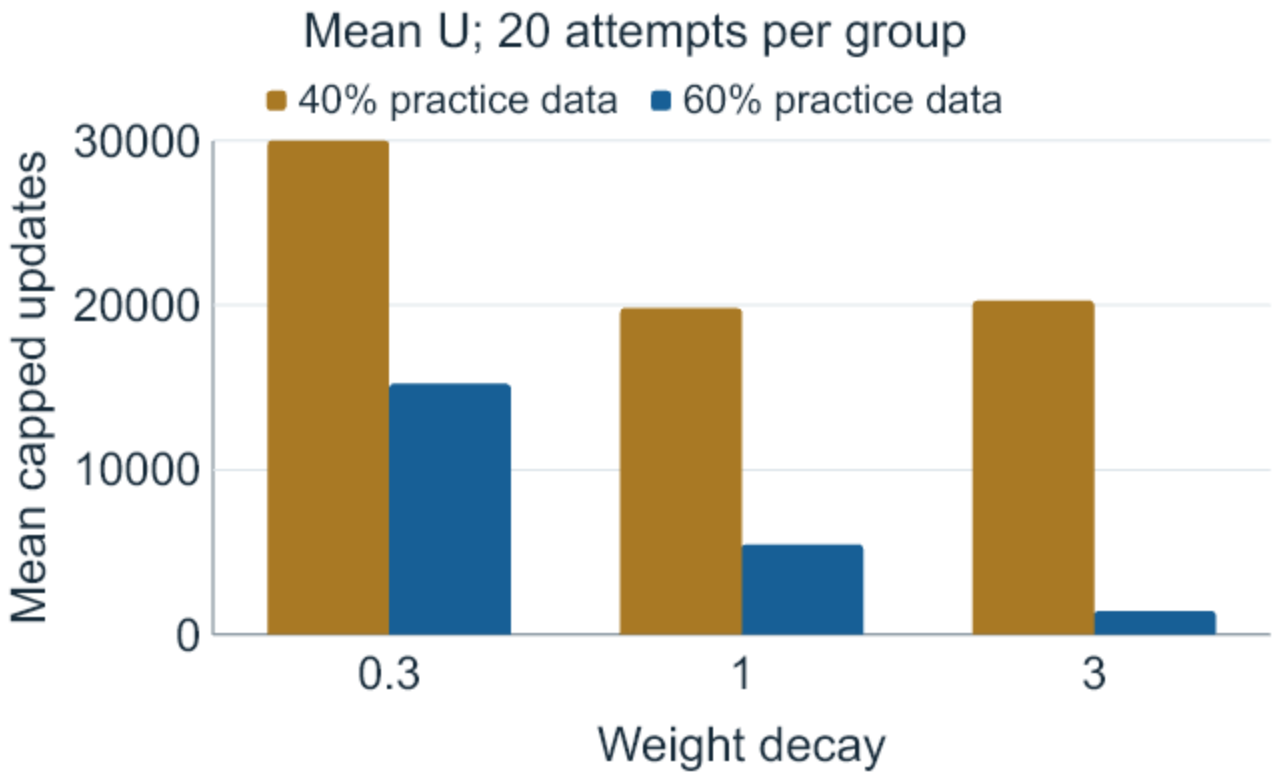
Practice accuracy rises before new-question accuracy



Practice threshold: 225 updates | First confirmed generalization T: 4,375 updates

Full results across the six settings

Average capped time U differs across the six settings

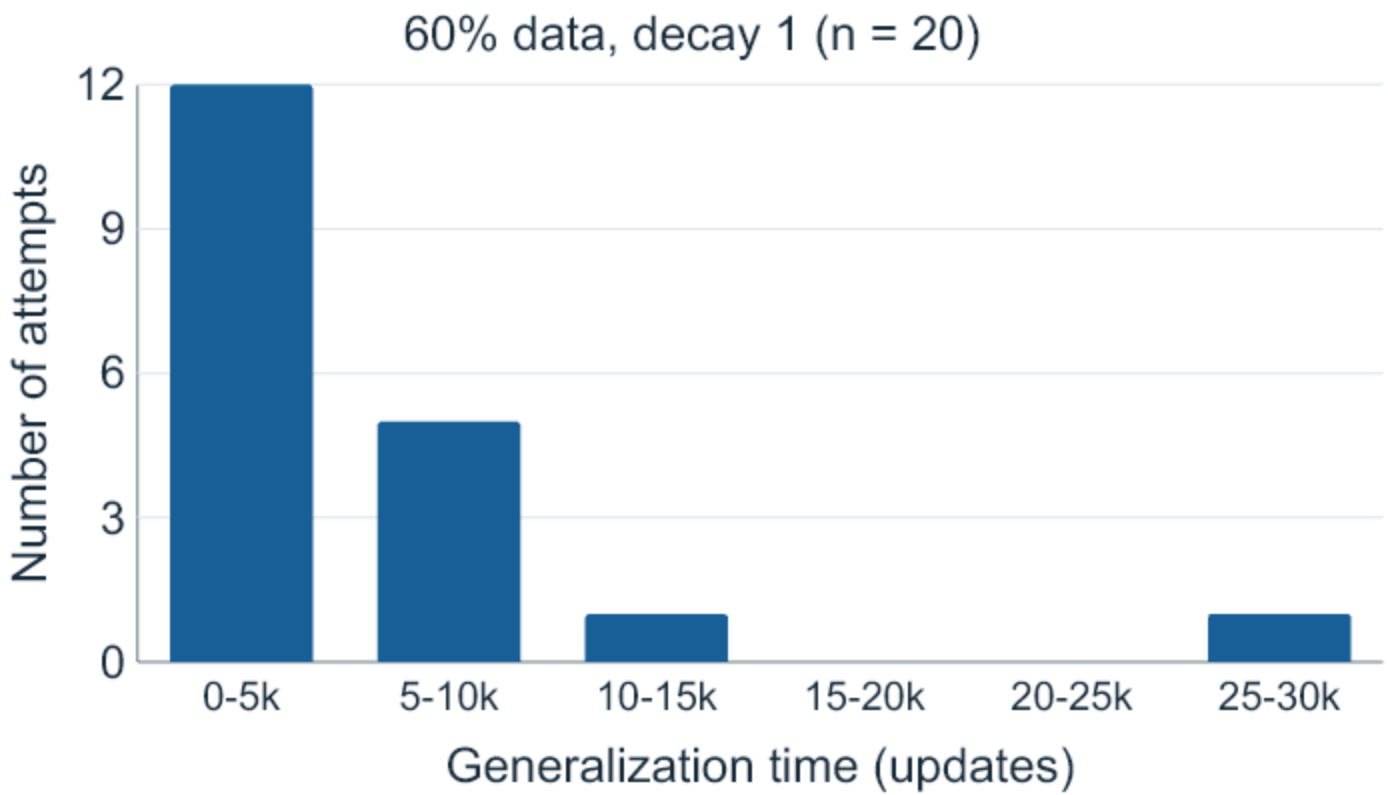


Data	Decay	Events	Mean U
40%	0.3	0/20	30,000
40%	1	14/20	19,823
40%	3	10/20	20,283
60%	0.3	19/20	15,235
60%	1	20/20	5,481
60%	3	20/20	1,430

An unfinished run is capped at C in U; its eventual T is still unknown.

Waiting time varies even at the reference setting

60% practice data, decay 1 | 20 attempts



Mean: 5,481 updates
Median: 4,025
SD: 5,548
Range: 1,875–27,175

Success: 20/20
Exact 95% CI: 83.2–100%
Bootstrap mean-U CI: 3,596–8,224

SD = standard deviation (spread). The exact interval reflects the small sample.

Which distribution describes the waiting times?

Three candidate patterns for the waiting-time data

Model fit accounts for the 25-update check grid and the 50-update confirmation lag.

Candidate pattern	AIC	Status
Exponential	257.15	Converged
Gamma	253.96	Converged
Weibull	256.85	Converged

AIC balances fit and model complexity; lower is better among these candidates.

The reference sample has 20 events and 0 unfinished attempts. Gamma ranks first here; this does not prove the true timing law.

Statistical comparisons and their limits

Statistics answer different questions; assumptions still matter

One-way ANOVA at 60% data: decay-group averages differ in this sample ($p < 0.001$).

Two-factor ANOVA: data fraction and decay differ; interaction is not clearly supported ($p = 0.289$).

Correlation: log decay and U move together at 60% data (Pearson −0.715; Spearman −0.866).

Regression: $R^2 = 0.630$ describes sample variation in capped time U.

Residuals fail a normal-shape check (Shapiro $p = 0.00020$); results are exploratory.

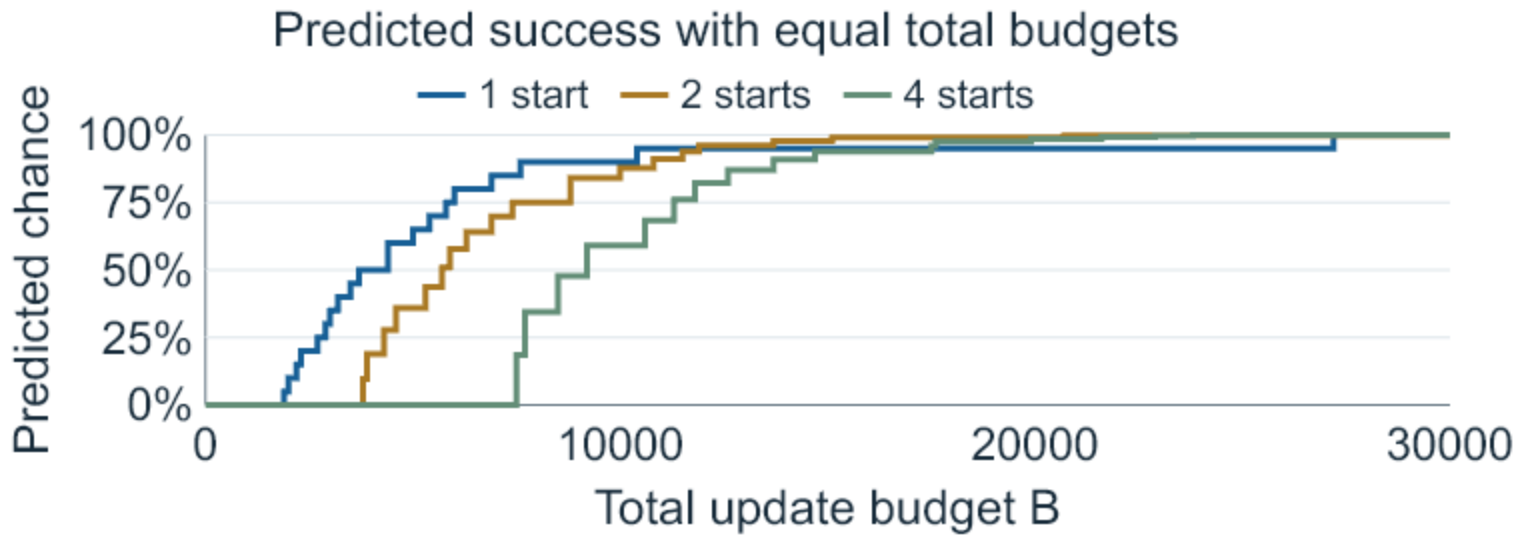
Permutation checks add a sensitivity view but assume exchangeable observations; they do not fix every model problem.

Predictions for one attempt versus restarts

Restart probability from the measured single-run curve

$P(\text{at least one success}) = 1 - [1 - F(B/k)]^k$

B = total updates | k = number of fresh starts | F(t) = chance one run succeeds by t



At B = 30,000:
1 start: 100%
2 starts: 99.8%
4 starts: 99.9%

From main data; these are predictions, not fresh-run evidence.

Fresh starting seeds test the predictions

Fresh seeds gave the same outcome at this generous budget

Policy	Allocation	Success	Exact 95% CI
1 start	1 × 30,000	10/10	69.2–100%
2 starts	2 × 15,000	10/10	69.2–100%
4 starts	4 × 7,500	10/10	69.2–100%

30 independent budget attempts; 70 individual training runs; exactly 30,000 logical updates per attempt.

No clear winner at this budget; equal results do not prove the policies equivalent. Runtime was not a controlled comparison.

What the completed study establishes

Findings and limits

Waiting time changes across random starts and training settings.

83/120 main runs confirmed generalization within 30,000 updates; 37 remained unfinished.

62/83 confirmed runs later fell below 95% held-out accuracy: first success is not permanent success.

All restart policies succeeded 10/10 at the tested budget, so the policy comparison is inconclusive.

Next research: smaller budgets, different practice/test splits, and sustained generalization.

The evidence is for this task, one fixed split and this small model.

Sources and proposed contribution

[1] Power et al. (2022)
Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets

https://arxiv.org/abs/2201.02177

[2] Truong et al. (2026), preprint
First-Passage Prediction of Grokking Delay: A Calibrated Law under AdamW with Causal Validation

https://arxiv.org/abs/2605.18845

[3] Streeter, Golovin and Smith (2007)
Restart Schedules for Ensembles of Problem Instances

https://www.cs.cmu.edu/~dgolovin/papers/aaai07b.pdf

[4] Official BAMAT207 syllabus | [5] This project's recorded final-study results

Proposed empirical connection: learning-time variation and equal-budget restarts. No worldwide-first claim is established.

Appendix A: terms and a worked probability

Words and a worked probability

Seed = repeatable random starting point. Update = one learning step.

Censored = the event was not seen before the observation limit.

Assume two independent starts each have a 60% chance; failure chance = 40%.

Both fail: 0.40 × 0.40 = 0.16. At least one succeeds: 1 − 0.16 = 0.84 (84%).

This calculation is hypothetical, not a measured result. Independence is an assumption.

Appendix B: statistical details for questions

Statistical details for questions

One-way ANOVA at 60% data: $p = 3.69 \times 10^{-10}$; permutation check $p \approx 0.001$.

Two-factor ANOVA: fraction $p \approx 5.65 \times 10^{-23}$; decay $p \approx 7.09 \times 10^{-12}$; interaction $p = 0.289$.

Regression on U: $R^2 = 0.630$; fitted 40%-to-60% difference ≈ −15,986 updates.

Policy intervals are wide because there are only ten attempts in each group.

The local copied-phrase screen is limited; an institutional plagiarism score is pending its chosen checker.

See the final report for exact methods, assumptions, uncertainty and source record.